Floating-Point, Fixed-Point, and Integer Number Representation Formats

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What Does This Code Print?

```
#include <iostream>
int main() {
  float a = 1000.43;
  float b = 1000.0;
  std::cout << a - b << '\n';
  return 0;}
```

- Answer: 0.429993
- Reason: 1000.43 is not exactly representable using float
- What if you change it from float to double?
- Answer: It correctly prints 0.43

https://isocpp.org/wiki/faq/newbie#floating-pt-errs

Does It Print TRUE(1) or FALSE(0)

```
#include <iostream>
  int main() {
   float p = 23.42;
   double q=23.42;
   bool isEqual = (p==q);
   std::cout<<isEqual<<"\n";}
```

- Answer: 0

Does it Print TRUE(1) or FALSE(0)

```
#include <iostream>
int main() {
  float p = 64;
  double q=64;
  bool isEqual = (p==q);
  std::cout<<isEqual<<"\n";}
```

- Answer: 1
- Reason: 23.42 is not stored exactly, whereas 64 is stored exactly. The error in storing 23.42 is different in float and double. I.e., a different approximation of 23.42 is stored in float and double; hence, they are unequal.

What Is the Output of This

```
#include<stdio.h>
int main() {
    float x = 0.1;
    if (x == 0.1)
        printf("IF");
    else if (x == 0.1f)
        printf("ELSE IF");
    else
        printf("ELSE");}
```

- Answer: ELSE IF

https://www.geeksforgeeks.org/comparison-float-value-c/

Reasoning

- float x = 0.1;
- printf("%d %d %d", sizeof(x), sizeof(0.1), sizeof(0.1f));
- This prints 4 8 4.
- The values used in an expression are considered as double unless a ‘f’ is specified at the end.
- 0.1 is not exactly represented by float

Limitations of IEEE-754 Floating-Point Format

Breaking algebraic rules. The floating-point formats may break algebraic rules during computation.

For instance, a floating-point addition is not always associative.

Let x = 1e30, y = -1e30 and z = 1

Now, (x+y)+z results in 1

x+(y+z) results in 0

```
#include<iostream>
int main() {
   float a = 1e30, b = -1e30, c = 1;
   float sum1 = (a+b)+c;
   float sum2 = a+(b+c);
   std::cout<<" sum1 "<<sum1<<" sum2 "<<sum2<<"\n";
   }
```

```
sum1 1 sum2 0
```